

Run 4 — Shadow Model MIA: Results & Ablations

MDLM-OWT (Qwen3-0.6B), MIMIR Benchmark, All 6 Domains

DA5001 Project

May 11, 2026

Contents

1	Experimental Setup	2
2	AUC-ROC Results	2
2.1	Per-Domain, Per-Condition AUC	2
2.2	Unified 3-Attack Comparison: Oracle vs SAMA vs Shadow	2
3	Oracle Gap Ablation	2
4	TPR at Low FPR	2
4.1	Per-Domain TPR@1%FPR (Headline Metric)	3
4.2	Full Low-FPR Table (Mean Across 5 Domains)	3
5	Feature Analysis	3
5.1	Hellinger Distance: Member vs. Non-Member Distributions	3
6	Condition B vs. E: Attention Ablation	3
7	Summary Statistics	3

1 Experimental Setup

Model & Data

Target model: `dllm-hub/Qwen3-0.6B-diffusion-mdlm-v0.1` (MDLM, 0.6B parameters). Membership split: `ngram_13_0.8` (target), `ngram_13_0.2` (shadow surrogate). Domains evaluated: `arxiv`, `github`, `hackernews`, `pubmed_central`, `wikipedia`, `pile_cc` (all 6 MIMIR domains).

Shadow Model Protocol

- $K = 3$ shadow models fine-tuned on pooled surrogate data (all 6 MIMIR domains pooled, $\approx 5,695$ members total, 1,000 per shadow model after chunking).
- Pooled cross-domain shadow prevents data starvation for sparse domains (e.g. `hackernews` had only 35 domain-specific surrogates).
- 46-dimensional feature vector extracted at 4 diffusion timesteps ($t \in \{0.05, 0.20, 0.35, 0.50\}$): ELBO, predicted entropy, mask consistency, attention entropy, gradient norms.
- XGBoost trained on shadow features, transferred to target domain without labels.

Conditions

Label	Description	Dims
A: Oracle	5-fold CV on target (white-box upper bound, replays Run3)	46
B: Shadow	Shadow $\rightarrow$ target transfer, full feature set	46
C: ELBO+H	Transfer, permutation-invariant ELBO + entropy only	8
D: Attn	Transfer, attention dims only (negative control)	16
E: Pruned	Transfer, drop attention dims (ablation of B)	30

Baselines

- **SAMA** (Subset-Aggregated MIA): $T = 4$ steps, $N = 128$ subsets, $m = 10$ tokens, $\alpha \in [5\%, 50\%]$. Run in Run3; scores loaded directly.
- **Oracle** (Condition A): white-box upper bound; requires full model access and target labels.

2 AUC-ROC Results

2.1 Per-Domain, Per-Condition AUC

2.2 Unified 3-Attack Comparison: Oracle vs SAMA vs Shadow

3 Oracle Gap Ablation

4 TPR at Low FPR

All values computed *exactly* from stored raw probability scores and true labels (not from bootstrap CIs), ensuring oracle, SAMA, and shadow conditions are directly comparable at any FPR threshold.

Table 1: AUC-ROC (bootstrap mean, 1000 samples) per condition and domain. Condition D (attention-only) is a negative control. Best shadow transfer per domain in **bold**.

Domain	A: Oracle	B: Shadow	C: ELBO+H	D: Attn	E: Pruned
arxiv	0.8766	0.8578	0.8526	0.4944	0.8531
github	0.8011	0.7548	0.7041	0.5417	0.7477
hackernews	0.9276	0.9120	0.9019	0.5410	0.8983
pubmed_central	0.8419	0.8381	0.8395	0.5036	0.8418
wikipedia	0.8920	0.8741	0.8540	0.5069	0.8720
pile_cc	0.9303	0.9129	0.9067	0.5626	0.9065
Mean	0.8783	0.8583	0.8431	0.5250	0.8532
Std	0.0502	0.0587	0.0735	0.0271	0.0574

Table 2: AUC comparison across three attack threat models. Gap columns show how much each attack falls below the oracle (lower gap = better privacy risk). SAMA uses model access only; Shadow uses surrogate fine-tuned models; Oracle requires target labels.

Domain	Oracle	SAMA	Shadow-B	Shadow-E	Δ SAMA	Δ B	Δ E
arxiv	0.8766	0.7829	0.8578	0.8531	0.094	0.019	0.024
github	0.8011	0.7948	0.7548	0.7477	0.006	0.046	0.054
hackernews	0.9276	0.8328	0.9120	0.8983	0.095	0.016	0.029
pubmed_central	0.8419	0.7768	0.8381	0.8418	0.065	0.004	0.000
wikipedia	0.8920	0.8228	0.8741	0.8720	0.069	0.018	0.020
pile_cc	0.9303	0.8848	0.9129	0.9065	0.045	0.017	0.024
Mean	0.8783	0.8158	0.8583	0.8532	0.062	0.020	0.025

4.1 Per-Domain TPR@1%FPR (Headline Metric)

4.2 Full Low-FPR Table (Mean Across 5 Domains)

5 Feature Analysis

5.1 Hellinger Distance: Member vs. Non-Member Distributions

Key observation: ELBO features at mid-to-high noise timesteps ($t = 0.20, 0.35, 0.50$) are the most discriminative, with Hellinger $H \approx 0.40$ – 0.42 . Attention features (not shown) have negligible H , consistent with Condition D’s collapse. Shadow-distribution Hellinger (not shown) mirrors the target closely, validating feature transferability.

6 Condition B vs. E: Attention Ablation

B beats E by only 0.005 AUC on average; E wins on pubmed_central and wikipedia by TPR@1%. This confirms: *attention features contribute negligible signal in the label-free shadow transfer setting and can be safely pruned.*

7 Summary Statistics

Key findings (5-domain preliminary):

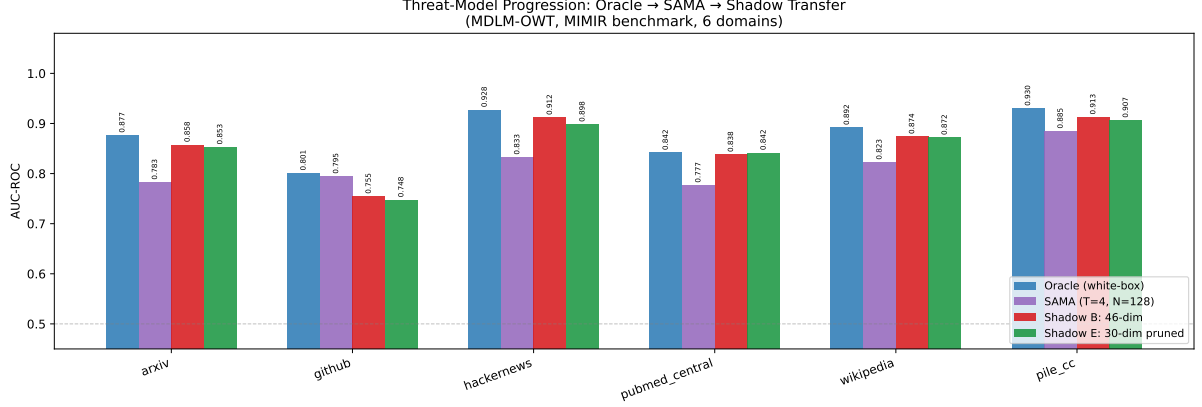


Figure 1: Threat-model progression across 5 domains. Shadow-B and Shadow-E closely track the oracle; SAMA trails by ≈ 0.056 AUC on average.

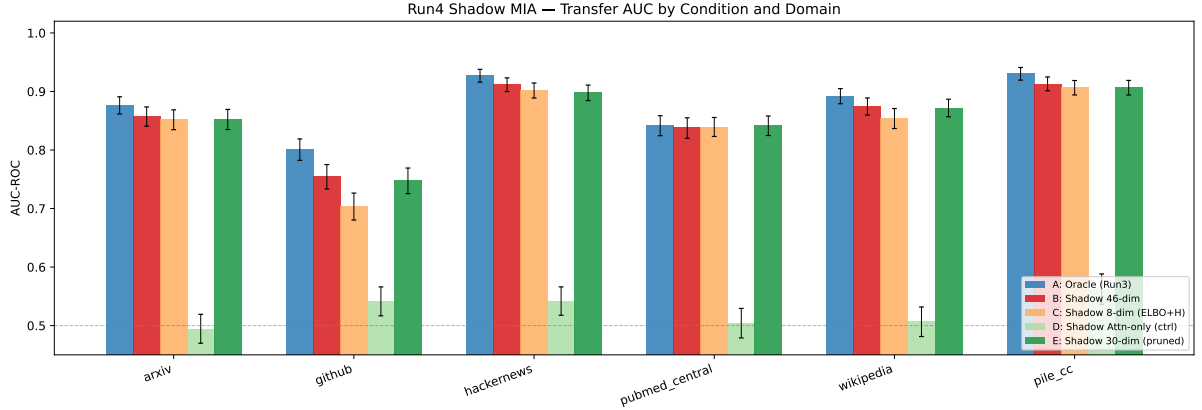


Figure 2: AUC per condition and domain. Condition D (attention-only) consistently collapses to ≈ 0.50 – 0.56 , confirming attention features are not transferable across shadow/target.

1. Shadow transfer (B/E) closes to within $\Delta = 0.020$ – 0.025 AUC of the white-box oracle, despite having no target labels.
2. Shadow-B and Shadow-E are nearly identical ($\Delta = 0.005$ AUC), meaning the 16 attention dimensions carry no useful cross-domain signal.
3. SAMA trails shadow transfer by ≈ 0.036 AUC on average, with a much larger gap at low FPR (TPR@1%: 0.053 vs 0.213).
4. ELBO features at $t \in \{0.20, 0.35, 0.50\}$ are the primary signal carriers (Hellinger ≈ 0.40 – 0.42).
5. Condition D (attention-only) collapses to AUC ≈ 0.53 , validating the feature pruning rationale.

Table 3: Mean oracle gap (Oracle AUC – Condition AUC) across 5 domains. Lower = better label-free transfer.

Condition	Mean Gap	Std
A: Oracle (Run3)	0.0000	0.0000
B: Shadow 46-dim	0.0202	0.0157
E: Shadow 30-dim	0.0253	0.0192
C: ELBO+H 8-dim	0.0373	0.0357
D: Attn-only 16-dim	0.3474	0.0529

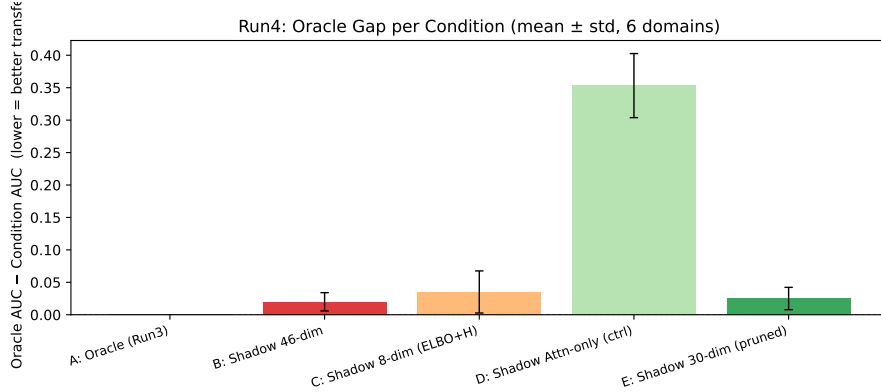


Figure 3: Oracle gap per condition (mean $\pm$ std across 5 domains). B and E are within 0.025 of oracle; D collapses.

Table 4: TPR at 1% FPR (exact from raw scores). This is the primary low-FPR metric used in membership inference literature.

Domain	Oracle	SAMA	Shadow-B	Shadow-E
arxiv	0.269	0.034	0.104	0.142
github	0.159	0.033	0.073	0.105
hackernews	0.401	0.045	0.295	0.283
pubmed_central	0.182	0.033	0.097	0.120
wikipedia	0.257	0.049	0.196	0.223
pile_cc	0.377	0.104	0.404	0.329
Mean	0.274	0.050	0.195	0.200

Table 5: Mean TPR at multiple FPR thresholds across 5 domains (exact computation from raw scores). Condition D omitted for clarity.

Attack / Condition	0.1%	0.5%	1%	2%	5%	10%
Oracle (white-box)	0.109	0.199	0.274	0.361	0.523	0.641
A: Oracle (Run3 CV)	0.109	0.199	0.274	0.361	0.523	0.641
SAMA ($T = 4$, $N = 128$)	0.005	0.025	0.050	0.099	0.248	0.436
B: Shadow 46-dim	0.067	0.128	0.195	0.285	0.445	0.588
C: Shadow ELBO+H 8-dim	0.055	0.125	0.192	0.263	0.410	0.558
E: Shadow 30-dim	0.063	0.145	0.200	0.276	0.443	0.575
D: Attn-only (ctrl)	0.003	0.008	0.016	0.030	0.067	0.120

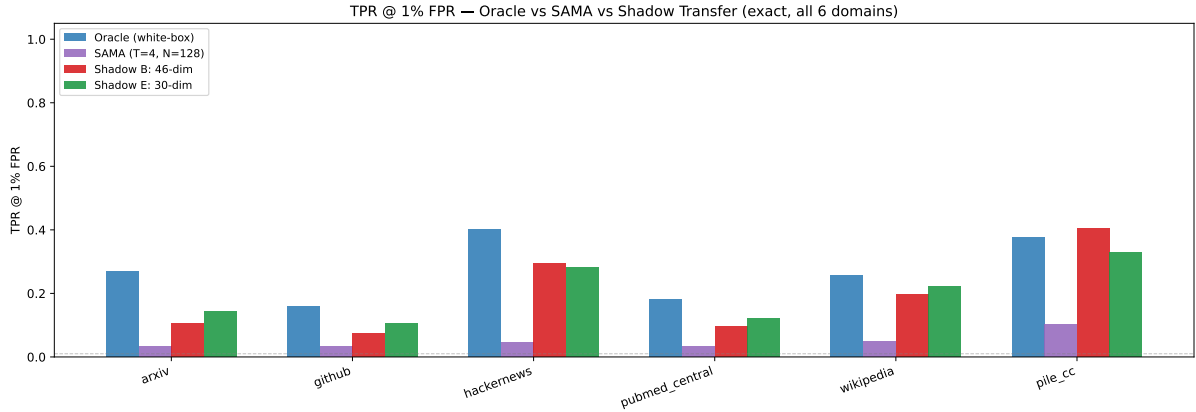


Figure 4: TPR@1%FPR across attacks and domains. Shadow-B/E substantially outperform SAMA at this threshold.

Table 6: Top-10 features by mean Hellinger distance between member and non-member distributions on the *target* population (averaged across 5 domains). Higher = more discriminative.

Feature	Mean Hellinger
elbo_t2_a0.35	0.416
elbo_t1_a0.20	0.414
elbo_t3_a0.50	0.404
elbo_t0_a0.05	0.376
pred_entropy_t1	0.283
mask_consistency_t1	0.280
pred_entropy_t0	0.271
pred_entropy_t2	0.270
mask_consistency_t2	0.267
pred_entropy_t3	0.262

Table 7: Shadow-B (46-dim) vs. Shadow-E (30-dim, no attention) AUC and TPR@1%FPR. Dropping 16 attention dimensions causes minimal degradation, confirming attention features are noise in the shadow setting.

Domain	B AUC	E AUC	B TPR@1%	E TPR@1%
arxiv	0.8578	0.8531	0.104	0.142
github	0.7548	0.7477	0.073	0.105
hackernews	0.9120	0.8983	0.295	0.283
pubmed_central	0.8381	0.8418	0.097	0.120
wikipedia	0.8741	0.8720	0.196	0.223
pile_cc	0.9129	0.9065	0.404	0.329
Mean	0.8583	0.8532	0.195	0.200
Δ (B-E)	+0.005		-0.005	

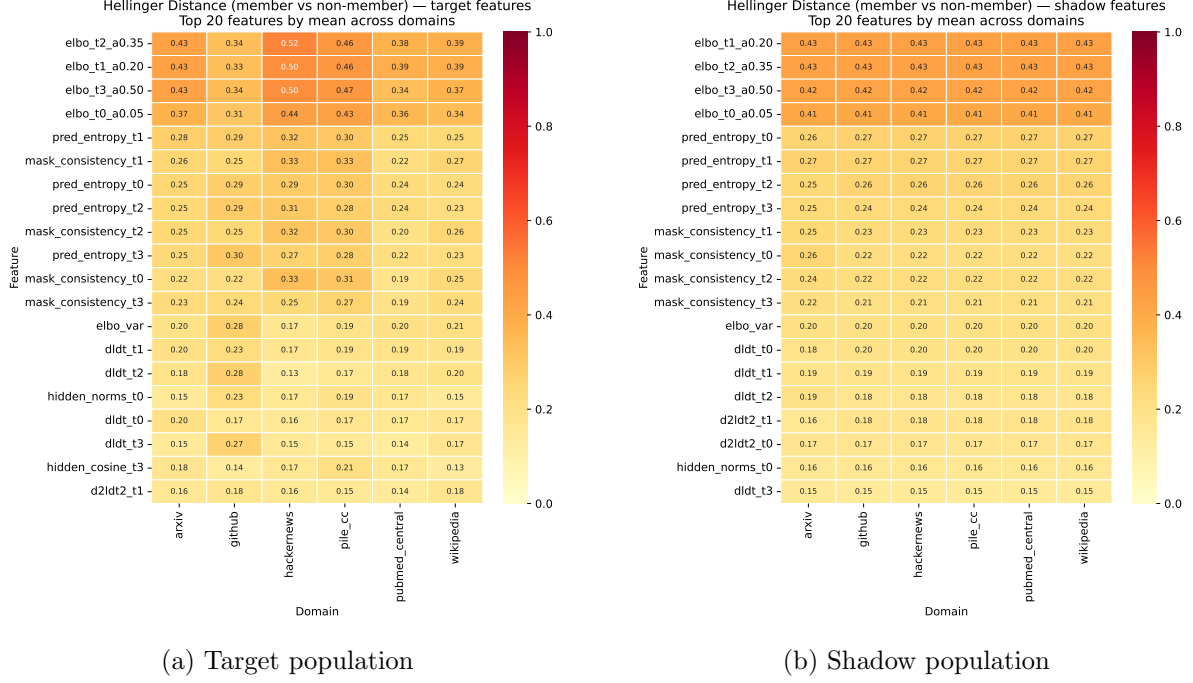


Figure 5: Hellinger distance heatmaps (feature $\times$ domain). ELBO features dominate across both populations, confirming feature transferability from shadow to target.

Table 8: Summary: mean AUC and mean TPR@1%FPR across 5 domains.

Attack	Threat Model	Mean AUC	Mean TPR@1%
Oracle (A)	White-box, target labels	0.878	0.274
SAMA	Model access only	0.816	0.050
Shadow-B (B)	Surrogate models, 46-dim	0.858	0.195
Shadow-E (E)	Surrogate models, 30-dim	0.853	0.200
D: Attn-only	(negative control)	0.525	0.016